Kevin Sabbe

https://sabbekosu.github.io/ | Corvallis, OR

Education

Oregon State University *Corvallis, OR | April 2025 - June 2027*
Master of Science - Robotics
Master of Science - Artificial Intelligence

Oregon State University *Corvallis, OR | March 2021 - June 2025*
Bachelor of Science, Double Major, Cum Laude
Computer Science - *Robotic Intelligence track*
Mechanical Engineering

Work Experience

Human-Robot Interaction Researcher *Corvallis, OR | May 2024 - Present*
OSU SHARE Lab (Robotics/AI)

- Engineered and optimized a real-time feedback system for a NAO robot, utilizing audio processing, computer vision, and LLMs to process audience reactions and enhance interactive performance.
- Developed a thermal imaging-based emotion detection system utilizing machine vision with OpenCV. Trained a machine learning model to classify user emotions using temperature data from facial regions.

Graduate Teaching Assistant *Corvallis, OR | September 2025 - Present*
OSU College of Engineering

- Provide instructions and personalized guidance on coding fundamentals for mechanical engineers and ROS 2 for new robotics graduate students.

Application Engineering Intern *Torrance, CA | June 2022 - September 2022*
Sanyo Denki America

- Created an embedded software controller for user and automated communication with multi-gantry systems, given user/camera sensor inputs. Prototyped mechanical systems for demonstrations.

Publications & Research

Text-Vision to Humanoid & General Robot Gesturing Model *(Github on website)*

- Implemented natural language-driven robot gesture generation, for any robot form factor, using a multi-modal LLM with feedback provided through simulated results and user-provided feedback.
- Robot agnostic platform tested to work with the NAO V6, Kinova Gen3, SAMI, and Spot robot.

Adaptive Querying for Reward Learning from Human Feedback *(Frontiers in Robotics & AI 2026)*

- Co-authored paper on adaptive feedback selection for learning robot reward functions from multiple modes of human feedback, balancing cost and information gain.

Audience Feedback Understanding in Robot Comedy (Under Review for Publication)

- Developed a real-time audio and visual feedback system for a NAO robot to provide real-time crowdwork
- Captures, processes, and classifies audio and visual feedback using machine learning models for computer vision optimized for low-latency performance on resource-constrained systems.

Leadership

Oregon State Men's Club Volleyball *Corvallis, OR | May 2022 - June 2025*
President

- Prime representative of the club.
  - Increased membership from 44 → 113
  - Fundraising increased from $1500/yr → $20,000/yr
  - Leader of a team of officers, and organized 5 tournaments/year, plus multiple clinics

Skills

Coding: Python, ReactJS, C++, Matlab, ROS/ROS2
Proficient in, CAD Software (Solidworks & Siemens NX), TensorFlow, Scikit-Learn